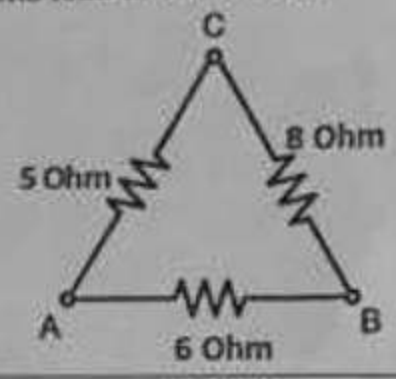
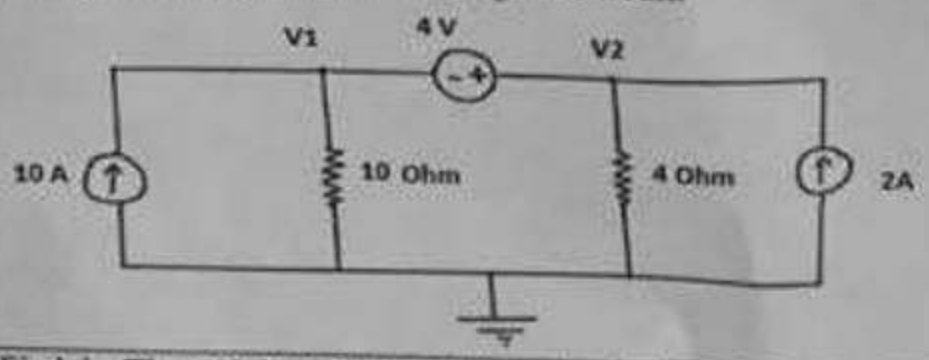
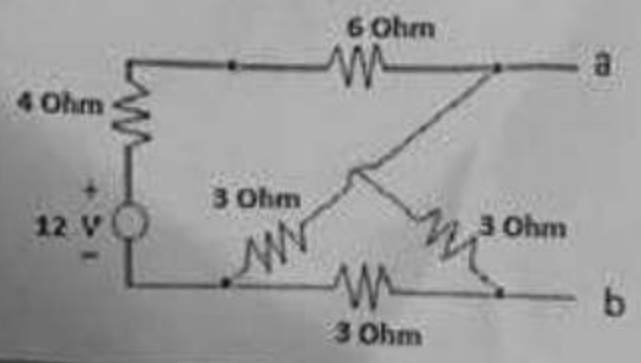
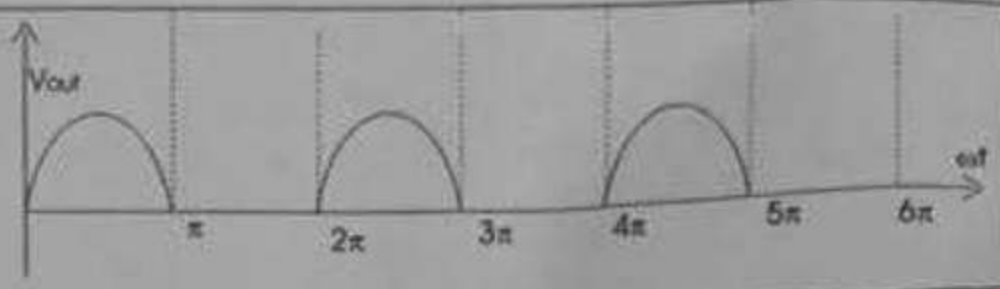
(Please write your Enrollment Number) Enrollment No. 00201012023

END TERM EXAMINATION
(CBCS)(SUBJECTIVE TYPE)(OFFLINE)
B. Tech<ECE-AI/ECE/CSE>, 1st Semester
(December, 2023)

Subject Code:BEC110	Subject: BASIC ELECTRICAL ENGINEERING
Time :3 Hours	Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

			(2.5*8 =20)	
Q1				
	(a)	Explain Kirchhoff's first and second law.		
	(b)	Convert the given Delta load into Star load 		
	(c)	Two impedances 4+3j Ohms and 4-3j Ohms are connected in parallel combination. This combination is then connected in series with 2+5j Ohm. Determine the equivalent impedance of the combination.		
	(d)	What do you understand by 3-phase connection? Give examples of it.		
	(e)	What is the significance of power factor of any circuit?		
	(f)	Why a moving coil instrument is not suitable for measuring ac quantities whereas a moving iron instrument can?		
	(g)	Why the transformer rating is in kVA and not in kW?		
	(h)	Explain Flemings left hand and right hand rule.		
		UNIT-I		
Q2	Find the value of V1 and V2 in the given circuit. 		(10)	
Q3	Find the Thevenin's equivalent of the circuit given in Figure. For what load across terminal ab, the circuit will deliver Maximum Power? Obtain Maximum Power Transfer. 		(10)	
		UNIT-II		
Q4	Derive the RMS and average value of half wave rectified sinusoidal voltage waveform shown in figure.		(10)	

			
Q5	A series RLC circuit undergoes resonance at a resonance frequency f_r. Find the resonance frequency, voltage across each component and current drawn by the circuit at resonance, if Voltage applied is 240V, R=80 ohms, L=10mH and C=10 micro F.	(10)	
	UNIT-III		
Q6	Explain the construction and working of Induction type instrument. Write its advantages and disadvantages over Moving iron instrument.	(10)	
Q7	Power in a three phase load is measured using 2 wattmeter method. (i) Draw the circuit diagram of connection of two wattmeters in 3 phase star load. (ii) Draw the phasor diagram and obtain the readings of two wattmeters (iii) Derive the expression of power factor of load in terms of two wattmeter readings.	(10)	
	UNIT-IV		
Q8	Draw the equivalent circuit and phasor diagram of a two winding transformer at a lagging power factor.	(10)	
Q9	Explain construction and working principle of a DC generator.	(10)	